

# Working Title

YT

September 28, 2026

## Abstract

**Introduction**  
**Methods**  
**Results**  
**Discussion**  
**Conclusion**

## 1 Introduction

the time of symptom onset determines access to treatment. (stroke treatment guidelines)  
study showed that unknown-onset stroke is common, distinct from wake-up stroke, and associated with lower reperfusion treatment, and poorer outcomes [1].

Main Aim:

The primary aim of this study was to examine temporal and geographic variation in uncertain symptom-onset stroke among stroke patients in Norway.

## 2 Methods

Data from the norwegian stroke registry, statistics norway

fom 2014 - tom 2013

inklusion criteria: I61 - hjerneblødning I63 - hjerneinfarkt I64 - hjerneslag, ikke spesifisert som infarkt eller blødning

nødvendig med sikre akutte fokale nevrologiske utfall

for bevistløse pasienter: akutte fokale nevrologiske utfall med positiv bildediagnostikk

innlagt innen 28 døgn etter symptomdebut (nytt slag etter 7 døgn registreres som nytt tilfelle, men ikke de innen 7 døgn)

ved trombolyse der det ikke blir sekvele eller bildefunn etter innleggels: kodes som I 63.

Pasienter med normal kontroll MR etter trombolytisk behandling skal kodes som hjerneinfarkt i ICD10.

Flow chart

88228

Innlagt i sykehus: 4571

83657

Våknet med symptom: 14757

68900

komnr missing: 38

68862

Table 1: Characteristics of patients

|                                    | <b>Uncertain onset<br/>(n = 12,612)</b> | <b>Other stroke<br/>(n = 55,233)</b> | <b>SMD</b> |
|------------------------------------|-----------------------------------------|--------------------------------------|------------|
| Age, mean (SD)                     | 73.89 (13.4)                            | 73.31 (13.4)                         | 0.043      |
| Female                             | 5,945 (47.1)                            | 25,337 (45.9)                        | 0.025      |
| Stroke type, Infarction            | 10,813 (85.7)                           | 47,360 (85.7)                        | 0.009      |
| Previous TIA                       | 880 (7.0)                               | 4,356 (7.9)                          | 0.043      |
| Previous MI                        | 1,482 (11.8)                            | 6,816 (12.3)                         | 0.020      |
| Atrial fibrillation                | 2,917 (23.1)                            | 13,119 (23.8)                        | 0.015      |
| Diabetes                           | 2,438 (19.3)                            | 9,234 (16.7)                         | 0.069      |
| Antihypertensive medication        | 6,508 (51.6)                            | 28,464 (51.5)                        | 0.040      |
| Lipid-lowering medication          | 3,700 (29.3)                            | 16,821 (30.5)                        | 0.050      |
| Thrombolysis                       | 161 (1.3)                               | 12,148 (22.0)                        | 0.685      |
| Smoking status                     |                                         |                                      | 0.086      |
| Never smoker                       | 4,673 (37.1)                            | 20,966 (38.0)                        |            |
| Current smoker                     | 2,717 (21.5)                            | 10,234 (18.5)                        |            |
| Former smoker                      | 2,998 (23.8)                            | 14,538 (26.3)                        |            |
| Unknown                            | 2,224 (17.6)                            | 9,495 (17.2)                         |            |
| Level of Consciousness (admission) |                                         |                                      | 0.087      |
| Alert                              | 10,856 (86.1)                           | 47,236 (85.5)                        |            |
| Mildly Drowsy                      | 800 (6.3)                               | 4,283 (7.8)                          |            |
| Markedly Drowsy                    | 349 (2.8)                               | 1,617 (2.9)                          |            |
| Unresponsive                       | 432 (3.4)                               | 1,699 (3.1)                          |            |
| Unknown                            | 175 (1.4)                               | 398 (0.7)                            |            |
| Pre-stroke modified Rankin Scale   |                                         |                                      | 0.098      |
| No symptoms                        | 6,884 (54.6)                            | 31,923 (57.8)                        |            |
| No significant disability          | 1,927 (15.3)                            | 8,805 (15.9)                         |            |
| Slight disability                  | 1,369 (10.9)                            | 5,319 (9.6)                          |            |
| Moderate disability                | 931 (7.4)                               | 3,829 (6.9)                          |            |
| Moderately severe disability       | 385 (3.1)                               | 1,612 (2.9)                          |            |
| Severe disability                  | 54 (0.4)                                | 227 (0.4)                            |            |
| Unknown                            | 1,062 (8.4)                             | 3,518 (6.4)                          |            |
| Living situation                   |                                         |                                      | 0.173      |
| Living with others                 | 6,192 (49.1)                            | 31,614 (57.2)                        |            |
| Living alone                       | 5,701 (45.2)                            | 20,548 (37.2)                        |            |
| Institution/nursing home           | 376 (3.0)                               | 1,490 (2.7)                          |            |
| Unknown                            | 333 (2.8)                               | 1,581 (2.8)                          |            |
| Education                          |                                         |                                      | 0.055      |
| No education                       | 60 (0.5)                                | 203 (0.4)                            |            |
| Compulsory education               | 4,150 (33.2)                            | 17,149 (31.3)                        |            |
| Upper-secondary education          | 5,868 (46.9)                            | 25,743 (47.0)                        |            |
| Higher education                   | 2,419 (19.3)                            | 11,606 (21.2)                        |            |
| Unknown                            | 15 (0.1)                                | 60 (0.1)                             |            |
| Household income                   |                                         |                                      | 0.068      |
| Low                                | 3,325 (26.4)                            | 13,437 (24.3)                        |            |
| M. low                             | 3,138 (24.9)                            | 13,278 (24.0)                        |            |
| M. high                            | 3,009 (23.9)                            | 13,470 (24.4)                        |            |
| High                               | 3,092 (24.5)                            | 14,885 (26.9)                        |            |
| Unknown                            | 48 (0.4)                                | 163 (0.3)                            |            |

SMD = standardized mean difference; SD = standard deviation; TIA = transient ischaemic attack.

## **2.1 Statistical analysis**

## **3 Results**

The study population included 67,845 patients with stroke registered in the Norwegian Stroke Registry between 2014 and 2023, of whom 12,612 (18.6%) had uncertain-onset stroke (table 1). Most patient characteristics were similar between patients with uncertain-onset stroke and all other strokes, with standardized mean differences generally below 0.10. The largest differences were observed for living situation, and thrombolysis, while differences in demographic characteristics, stroke type, and cardiovascular history were modest.

### **Temporal pattern**

### **Geographic patterns**

## unadjusted and unadjusted pairs from 2014 to 2023

Table 2: Fixed effect estimates with and without BYM2 spatial effect. Values are log-odds coefficients with 95% credible intervals. Bold indicates intervals excluding zero.

| Variable                                                                   | Without BYM2 $\beta$ (95% CrI) | With BYM2 $\beta$ (95% CrI)    | $\Delta\beta$ |
|----------------------------------------------------------------------------|--------------------------------|--------------------------------|---------------|
| (Intercept)                                                                | <b>-1.701 (-1.924, -1.478)</b> | <b>-1.834 (-2.065, -1.603)</b> | 0.133         |
| Bosituasjon: bodde alene                                                   | <b>0.408 (0.366, 0.450)</b>    | <b>0.428 (0.385, 0.471)</b>    | -0.020        |
| Bosituasjon: bodde i institusjon/sykehjem                                  | <b>0.321 (0.216, 0.425)</b>    | <b>0.349 (0.242, 0.457)</b>    | -0.028        |
| Bevissthetsgrad: døsig, reagerer adekvat ved lett stimulering              | <b>-0.202 (-0.280, -0.125)</b> | <b>-0.174 (-0.254, -0.095)</b> | -0.028        |
| Bevissthetsgrad: døsig, reagerer først ved kraftigere/gjentatt stimulering | -0.079 (-0.192, 0.035)         | -0.027 (-0.144, 0.089)         | -0.051        |
| Bevissthetsgrad: reagerer ikke / ikke-måltrettet bevegelse                 | 0.073 (-0.031, 0.177)          | <b>0.128 (0.020, 0.235)</b>    | -0.055        |
| Røyking: røyker                                                            | <b>0.153 (0.096, 0.210)</b>    | <b>0.148 (0.090, 0.207)</b>    | 0.005         |
| Røyking: eks-røyker                                                        | -0.041 (-0.092, 0.010)         | -0.011 (-0.064, 0.042)         | -0.030        |
| Kjønn: ErMann = nei                                                        | <b>-0.074 (-0.117, -0.031)</b> | <b>-0.070 (-0.115, -0.026)</b> | -0.004        |
| Utdanning: obligatorisk utdanning                                          | -0.075 (-0.247, 0.097)         | -0.062 (-0.239, 0.115)         | -0.013        |
| Utdanning: mellomutdanning                                                 | -0.107 (-0.278, 0.065)         | -0.107 (-0.283, 0.070)         | -0.000        |
| Utdanning: universitets- og høyskoleutdanning                              | -0.153 (-0.330, 0.024)         | -0.124 (-0.306, 0.057)         | -0.029        |
| Inntekt: samlet kvartil 2                                                  | <b>-0.056 (-0.110, -0.002)</b> | <b>-0.057 (-0.113, -0.001)</b> | 0.001         |
| Inntekt: samlet kvartil 3                                                  | <b>-0.089 (-0.147, -0.032)</b> | <b>-0.081 (-0.140, -0.021)</b> | -0.009        |
| Inntekt: samlet kvartil 4                                                  | <b>-0.152 (-0.217, -0.087)</b> | <b>-0.130 (-0.197, -0.063)</b> | -0.022        |
| Inntekt: samlet ukjent/9                                                   | 0.125 (-0.196, 0.445)          | 0.141 (-0.188, 0.470)          | -0.016        |
| Tidligere hjerneslag: nei                                                  | -0.032 (-0.082, 0.017)         | -0.030 (-0.081, 0.021)         | -0.002        |
| Tidligere TIA: nei                                                         | <b>0.110 (0.041, 0.179)</b>    | <b>0.099 (0.028, 0.170)</b>    | 0.011         |
| Tidligere hjerteinfarkt: nei                                               | 0.060 (-0.001, 0.121)          | 0.042 (-0.022, 0.105)          | 0.019         |
| Atrieflimmer: nei                                                          | <b>0.049 (0.002, 0.097)</b>    | 0.040 (-0.009, 0.089)          | 0.010         |
| Diabetes: nei                                                              | <b>-0.213 (-0.262, -0.161)</b> | <b>-0.232 (-0.283, -0.181)</b> | 0.019         |
| ASA pre: nei                                                               | <b>0.064 (0.017, 0.111)</b>    | <b>0.053 (0.005, 0.101)</b>    | 0.011         |
| Statin/lipid pre: nei                                                      | <b>0.049 (0.002, 0.095)</b>    | <b>0.065 (0.017, 0.113)</b>    | -0.016        |
| Dipyridamol pre: nei                                                       | <b>0.167 (0.062, 0.273)</b>    | <b>0.131 (0.023, 0.240)</b>    | 0.036         |
| BT-medikasjon pre: nei                                                     | -0.030 (-0.072, 0.013)         | -0.006 (-0.050, 0.038)         | -0.023        |

## 4 Discussion

This is a placeholder Discussion section added to verify the LaTeX build and editor setup. Replace this text with interpretation of results and context.

## Supplementary

### References

- [1] M Søyland et al. “Wake-up stroke and unknown-onset stroke; occurrence and characteristics from the nationwide Norwegian Stroke Register”. In: *European Stroke Journal* 7.2 (2022), pp. 143–150. DOI: 10.1177/23969873221089800.







